

Xiang Ge (Xiangge) Luo

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Research interests

Probabilistic graphical models, stochastic processes, population dynamics, phylogenetics, evolutionary theory

Education

ETH Zurich

PHD IN COMPUTATIONAL BIOLOGY - ETH MEDAL FOR OUTSTANDING THESIS

- Thesis: Modeling tumor mutation trees for evolution-guided precision oncology
- Advisor: Prof. Dr. Niko Beerenwinkel | Second advisor: Dr. Jack Kuipers
- Doctoral committee: Prof. Dr. Ben Raphael, Prof. Dr. Jasmine Foo

Basel, Switzerland

2020.11 - 2025.12

ETH Zurich

MASTER OF SCIENCE IN STATISTICS

- GPA: 5.88/6 – Graduated with distinction
- Thesis: Learning Bayesian networks from ordinal data
- Advisor: Dr. Jack Kuipers

Zurich, Switzerland

2018.09 - 2020.09

University of Waterloo

BACHELOR OF MATHEMATICS

- GPA: 92% – Graduated with distinction, Dean's Honours List

Waterloo, Canada

2013.09 - 2018.04

Research Experience

Computational Biology Group, ETH Zurich

POSTDOCTORAL RESEARCHER

- Advisor: Prof. Dr. Niko Beerenwinkel
- Developing an evolutionary forecasting framework for treatment resistance based on multitype branching processes, integrating longitudinal single-cell phylogenies from 231 acute myeloid leukemia patients across diagnosis, remission, and relapse.
- Designing an interactive web tool to visualize patient-specific fitness landscapes and forecast survival outcomes under alternative therapeutic strategies.

Basel, Switzerland

2026.03 - present

Computational Biology Group, ETH Zurich

PHD RESEARCHER

- Advisor: Prof. Dr. Niko Beerenwinkel
- Developed mathematical and statistical models to infer cancer evolution from cross-sectional single-cell phylogenies, aiming to predict drug resistance and facilitate evolution-guided precision oncology.

Basel, Switzerland

2020.11 - 2026.02

Institute of Molecular Systems Biology, ETH Zurich

RESEARCH ASSISTANT

- Advisor: Prof. Dr. Mattia Zampieri
- Analyzed E. coli drug response metabolomics data to investigate regulatory and functional relationships between metabolites.

Zurich, Switzerland

2019.03 - 2019.07

Seminar for Statistics, ETH Zurich

STATISTICS LAB RESEARCHER

- Advisor: Dr. Markus Kalisch
- Analyzed mouse experimental data using mixed models to assess bone loss and identify early osteoporosis markers.

Zurich, Switzerland

2019.03 - 2019.06

Department of Statistics and Actuarial Science, University of Waterloo

UNDERGRADUATE RESEARCHER

- Advisor: Prof. Dr. Alexander Schied
- Derived the Nash equilibrium for risk-averse investors in a market impact game with transient price impact and quadratic transaction costs over finite and infinite horizons.

Waterloo, Canada

2017.09 - 2017.12

Publications

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Numerical approximations of population size distributions for multi-type branching processes. *Manuscript in preparation*, 2026.

Xiang Ge Luo, Jack Kuipers, Kevin Rupp, Koichi Takahashi and Niko Beerenwinkel. Bayesian inference of fitness landscapes via tree-structured branching processes. *Bioinformatics*, 41:i160-i169, 2025. (**ISMB/ECCB 2025**, **Best Paper Award**, acceptance rate 17%)

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees. *Nature Communications*, 14:3676, 2023. (**RECOMB 2022**, acceptance rate 21.3%; **SIB Remarkable Outputs 2023 Award**)

Xiang Ge Luo, Giusi Moffa and Jack Kuipers. Learning Bayesian networks from ordinal data. *Journal of Machine Learning Research*, 22:1-44, 2021.

Xiangge Luo and Alexander Schied. Nash equilibrium for risk-averse investors in a market impact game with transient price impact. *Market Microstructure and Liquidity*, 5:2050001, 2020.

Awards & Scholarships

2026	ETH Medal for outstanding doctoral thesis , ETH Zurich	Switzerland
2025	ISMB/ECCB 2025 Ian Lawson Van Toch Memorial Award for Best Student Paper , ISCB	United Kingdom
2025	ISMB/ECCB 2025 Conference Fellowship & Dependent Care Grant , ISCB (USD 950)	United Kingdom
2024	SIB Remarkable Outputs 2023 Award , Swiss Institute of Bioinformatics	Switzerland
2018	Undergraduate Student Research Award , NSERC (CAD 4,500)	Canada
2018	President's Research Award , University of Waterloo (CAD 1,500)	Canada
2016	President's International Experience Award , University of Waterloo (CAD 1,500)	Canada
2014	Isabel Farrar Undergraduate Entrance Scholarship , University of Waterloo (CAD 4,000)	Canada
2014	President's Scholarship of Distinction , University of Waterloo (CAD 2,000)	Canada

Conferences, Seminars & Workshops

2025.07	Contributed talk , ISMB/ECCB 2025 Conference	Liverpool, UK
	• Bayesian inference of fitness landscapes via tree-structured branching processes	
2025.04	Poster , 6th Zurich Precision Oncology Symposium	Zurich, Switzerland
	• FiTree: fitness inference from single-cell phylogenies applied to acute myeloid leukemia data	
2025.03	Invited talk , D-BSSE Departmental Seminar Series, ETH Zurich	Basel, Switzerland
	• Fitness inference from single-cell phylogenies using tree-structured branching processes	
2025.02	Contributed talk , PhyloBasel Seminar Series, ETH Zurich	Basel, Switzerland
	• Bayesian inference of fitness landscapes via tree-structured branching processes	
2024.06	Award talk , SIB Days – The Swiss Bioinformatics Summit	Virtual
	• Mining tumor mutation trees with TreeMHN for evolution-guided precision oncology	
2023.06	Contributed talk , SKINTEGRITY.CH Annual Retreat	Spiez, Switzerland
	• Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	
2022.06	Contributed talk , Mutual Hazard Networks Workshop	Virtual
	• Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	
2022.05	Contributed talk , RECOMB 2022 Conference	San Diego, USA
	• Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	
2022.04	Contributed talk , OLISSIPO Exchange Week	Lisbon, Portugal
	• Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	
2022.03	Poster , Ascona Workshop 2022	Ascona, Switzerland
	• Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	

Teaching

Spring 2023	Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich
Spring 2022	Head Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich
Spring 2022	Teaching Assistant , LSZGS Advanced Block Course Computational Biology	ETH Zurich
Spring 2021	Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich

Supervision ---

- 2025 **Marie Dogo**, Master Student, Mines Paris - PSL
- 2023 **Laura Quintas**, Master Student, INESC-ID (Co-supervised with Monica Baciu-Dragan)
- 2023 **Laurenz Keller**, Bachelor Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)
- 2022 **Jiayi Wang**, Master Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)

Academic Services ---

Journal Reviewer for Genome Research, Bioinformatics **Conference Reviewer** for RECOMB, ISMB

Languages and Skills ---

Coding Python, R, C++, MATLAB, Git, Snakemake, bash, $\text{\LaTeX}$
Languages English (working proficiency), Mandarin (native), Cantonese (native)